Sujal Lohar

Full-Stack Software Engineer & DevSecOps Architect

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SUMMARY

Software Engineer and DevSecOps Architect with core competencies in distributed microservices (Java 21, Spring Boot 3.2, gRPC, PostgreSQL, Redis), container security, and cryptographic audit ledgers. Experienced in designing Saga choreography state machines, multi-stage non-root Docker deployments, and automated CI/CD security quality gates.

EDUCATION

Institute of Technology, Nirma University

Aug 2023 – May 2027

Bachelor of Technology (B.Tech) in Computer Science and Engineering (Minor in Cybersecurity) — ***8.44 / 10.0 CGPA***

- Core Coursework: Distributed Systems, Operating Systems, Computer Networks, Data Structures & Algorithms, Database Management Systems, Object-Oriented Programming (Java/C++), Cryptography & Network Security, Cloud Computing.
- Recipient: Ministry of External Affairs SPDC Full Scholarship (2023 - 2027).
- Competitive Programming: Codeforces Specialist (Rating 1400+), solve algorithmic graph and dynamic programming problems daily.

EXPERIENCE

Undergraduate Research Fellow | Institute of Technology, Nirma University

Aug 2024 – Present

Ahmedabad, Gujarat, India

- Engineered decentralized federated learning pipelines using FedProx across 40,000+ MIMIC-III ICU records.
- Attained 96.4% ICU hazard classification accuracy with multi-hop attention rollout for clinical interpretability.
- Co-authored 5 peer-reviewed papers indexed in IEEE Xplore and presented at international conferences.

Automation & Web Developer Intern | T.I.M.E. (Triumphant Institute of Management Education)

Jun 2025 – Jul 2025

Ahmedabad, Gujarat, India

- Constructed containerized Python dispatch service using Twilio WhatsApp API and MongoDB, increasing delivery speed by 30%.
- Eliminated 10+ hours per week of manual operational overhead for administrative staff.

Web Development Intern | The Akshaya Patra Foundation

Jun 2024 – Jul 2024

Ahmedabad, Gujarat, India

- Engineered operational web app on Node.js, Express, and MySQL replacing paper logs across 2 industrial kitchens.
- Reduced reporting latency and saved 10+ hours per week in manual kitchen reconciliation.

Selected Delegate (HPAIR Tokyo) | Harvard Project for Asian and International Relations

Aug 2025 – Oct 2026

Tokyo, Japan

- Selected among international applicants (sub-10% acceptance rate) to represent India and Nirma University at Tokyo summit.
- Collaborated on technology policy and public infrastructure security frameworks.

SELECTED ENGINEERING PROJECTS

GovAsset Nexus: Polyglot Microservices Platform

Oct 2026

Stack: Java 21 • Spring Boot 3.2 • Redis 7.2 • gRPC • PostgreSQL • MongoDB • Kafka • Docker • Kubernetes • Istio

- Architected 5 polyglot microservices on Java 21 / Spring Boot 3.2 and Node.js using gRPC and Kafka.
- Engineered distributed Saga choreography with PostgreSQL transactional outbox for zero-loss recovery.
- Deployed to Kubernetes with Istio service mesh and Redis 7.2 sliding-window rate limiting.

GovAsset 360: Cryptographic WORM & AppSec Gateway

Sep 2026

Stack: Node.js • TypeScript • Web Crypto SHA-256 • Firestore WORM • Docker • Trivy • GitLeaks • GitHub Actions • Zod

- Engineered SHA-256 cryptographic hash-chained audit engine with WORM append-only enforcement.
- Constructed hardened AppSec Express gateway with Zod schema validation and sliding-window rate limiters.
- Built multi-stage non-root Docker builds verified via Trivy and GitLeaks CI/CD security gates.

Federated Clinical Predictor & XAI Engine

Aug 2025

Stack: Python 3.11 • PyTorch • FedAvg / FedProx • TabTransformer • MIMIC-III Dataset • Scikit-Learn • Flask • Docker

- Formulated decentralized federated learning pipeline using FedProx on 40,000+ MIMIC-III ICU records.
- Attained 96.4% predictive accuracy with multi-hop attention rollout for clinical feature interpretability.
- Co-authored research paper published in IEEE Xplore and presented at international conferences.

RESEARCH PUBLICATIONS

"Federated Transformer-Based Learning for Predicting ICU Events with Multi-hop XAI Explanation Models" — IEEE Conference (Published in IEEE Xplore) (IEEE, 2025) [Published]

TECHNICAL SKILLS

Programming Languages: C++, Python, JavaScript / TypeScript, Java, SQL, Bash / Shell

Cloud, DevOps & DevSecOps: Docker, Kubernetes, GitHub Actions, DevSecOps (Trivy / GitLeaks), AWS & OCI, Linux System Administration

Backend & Distributed Systems: Spring Boot 3.2, Node.js & Express, NestJS, gRPC & Protocol Buffers, Kafka & Redis

Databases & Storage: PostgreSQL, MongoDB, Redis, Firebase Firestore

CERTIFICATIONS & HONORS

Oracle Cloud Infrastructure 2024 DevOps Professional (Oracle Cloud Infrastructure, Oct 2024)

DevOps Engineering Job Simulation (Macquarie Group (via Forage), May 2024)

Selected Delegate - HPAIR Harvard Conference (Harvard College Project for Asian and International Relations (Tokyo, Japan))

Oracle Cloud Infrastructure 2024 Generative AI Certified Professional (Oracle Cloud Infrastructure, Aug 2024)

Top 8 Sanchar Mitras State Recognition (Department of Telecommunications (DoT), Government of India - Gujarat LSA)

Scholarship Programme for Diaspora Children (SPDC) (Ministry of External Affairs, Government of India)